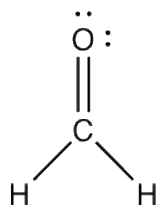


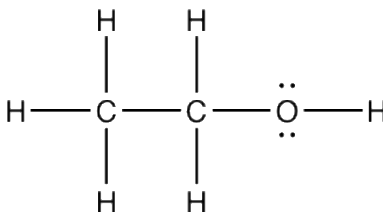
1. Which molecule is most likely to exhibit resonance?

- A)  $\text{CO}_2$
- B)  $\text{N}_2$
- C)  $\text{O}_3$
- D)  $\text{XeF}_4$
- E)  $\text{CO}$

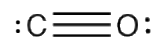
2. Which statement is true regarding the carbon to oxygen bonds in the three molecules shown below?



Structure I



Structure II



Structure III

- A) Structure II has the strongest bond.
- B) Structure III has the longest bond.
- C) Structure I has the weakest bond.
- D) Structure II has the shortest bond.
- E) Structure III has the strongest bond.

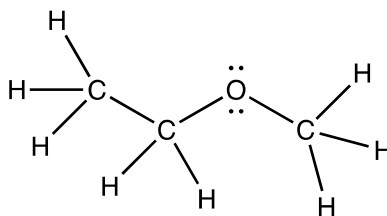
3. Which molecule or molecules does not have an ideal bond angle of  $90^\circ$ ?

- I.  $\text{SCl}_4$
- II.  $\text{ClF}_3$
- III.  $\text{CCl}_4$

- A) I only
- B) II only
- C) Both I and III
- D) Both I and II
- E) All of I, II, and III

4. Which statement is true regarding molecule polarity?
- A) Molecules with lone pairs on the central atom must be polar.
  - B)  $\text{CH}_2\text{F}_2$  could be polar or nonpolar.
  - C) Molecules composed only of nonpolar bonds may be either polar or nonpolar.
  - D) A square planar molecule could be polar or nonpolar.
  - E) Molecules in which the "outer" atoms are not all the same must be polar.
5.  $\text{H}_2\text{O}$  and  $\text{SO}_2$  both have bent molecular geometries (shapes). Which statement is true regarding the two molecules?
- A) The molecules have the same electron geometries.
  - B) The H-O-H bond angle is less than the O-S-O bond angle.
  - C) Both molecules can be classified as tetrahedral.
  - D) The two molecules have identical bond angles.
  - E) Both statements A and C are true.
6. Which combination(s) of orbitals can be used to form sigma bonds?
- I. An  $s$  orbital and an  $sp^3$  hybrid orbital
  - II. An  $sp^2$  hybrid orbital and an  $sp^2$  hybrid orbital
  - III. An  $sp^2$  orbital and a  $p$  orbital oriented along the internuclear axis
- A) I only
  - B) II only
  - C) Both II and III
  - D) All of I, II, and III
  - E) Both I and II
7. Which answer options describes the triple bond in ethyne?
- $$\text{H} - \text{C} \equiv \text{C} - \text{H}$$
- A) A sigma bond formed between  $sp$  hybrid orbitals on both C atoms and two pi bonds formed between unhybridized  $p$  orbitals on both C atoms
  - B) Three sigma bonds formed between unhybridized  $p$  orbitals on both C atoms
  - C) A sigma bond formed between unhybridized  $p$  orbitals on both C atoms and two pi bonds formed between unhybridized  $p$  orbitals on both C atoms
  - D) Two sigma bonds formed between unhybridized  $s$  orbitals on both C atoms and pi orbital formed between  $sp$  hybrid orbitals on both C atoms
  - E) A sigma bond formed between  $sp^3$  hybrid orbitals on both C atoms and two pi bonds, one formed between  $sp^3$  hybrid orbitals and the other from unhybridized  $p$  orbitals on both C atoms.

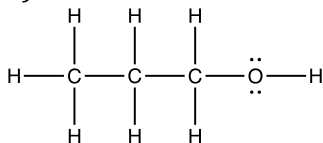
8. What is the *primary* intermolecular force between molecules of methoxyethane?



- A) Ion-dipole interactions  
B) Ion-ion interactions  
C) Hydrogen bonding interactions  
D) Dipole-dipole interactions  
E) London dispersion forces
9. Which two molecules can participate in hydrogen bonding interactions with one another?
- I.  $\text{CH}_2\text{F}_2$  and  $\text{H}_2\text{O}$   
II.  $\text{NH}_3$  and  $\text{H}_2\text{O}$   
III. Methoxyethane and methoxyethane (see question 8 for structure)
- A) I only  
B) II only  
C) Both I and II  
D) Both II and III  
E) All of I, II, and III
10. Which statement is true regarding vapor pressure and boiling point?
- A) Substances with higher vapor pressures have higher boiling points.  
B) As atmospheric pressure decreases, so does boiling point.  
C) As atmospheric pressure increases, so does normal boiling point.  
D) When intermolecular forces are weaker, it is harder to get the vapor pressure to be equal to the atmospheric pressure so boiling point is higher.  
E) Boiling points are higher for substances that evaporate more readily.

11. Which molecule has the *second* lowest boiling point?

- A)  $\text{CaO(s)}$
- B)  $\text{NaCl(s)}$
- C)  $\text{CCl}_4$
- D)  $\text{CH}_4$
- E)



12. Which statement is true when silicon is doped with phosphorus?

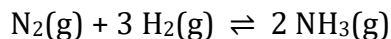
- A) Electrons are added to create a donor band that is closer in energy to the conduction band, thus increasing conductivity.
- B) Electrons are removed to create an acceptor band that is farther in energy from the conduction band, thus decreasing conductivity.
- C) Electrons are added to create a donor band that is farther in energy from the conduction band, thus decreasing conductivity.
- D) Electrons are removed to create an acceptor band that is closer in energy to the conduction band, thus increasing conductivity.
- E) There is no change in conductivity.

13. The total change in enthalpy is 30.1 kJ when 50.0 g of an unknown substance is heated from 92.5 to 145 K at constant pressure. Given the information regarding the unknown substance below, determine the specific heat of the liquid phase of the unknown.

Molar mass = 16.05 g/mol  
Boiling point = 111.7 K  
Freezing point = 90.7 K  
 $\Delta H_{\text{fus}} = 58.6 \text{ J/g}$   
 $\Delta H_{\text{vap}} = 511 \text{ J/g}$   
 $C_{\text{s (gas)}} = 2.09 \text{ J/g}\cdot^\circ\text{C}$   
 $C_{\text{s (solid)}} = 1.03 \text{ J/g}\cdot^\circ\text{C}$

- A)  $1.11 \text{ J/g}\cdot^\circ\text{C}$
- B)  $1.56 \text{ J/g}\cdot^\circ\text{C}$
- C)  $3.48 \text{ J/g}\cdot^\circ\text{C}$
- D)  $9.60 \text{ J/g}\cdot^\circ\text{C}$
- E)  $10.8 \text{ J/g}\cdot^\circ\text{C}$

14. If the rate of formation of  $\text{NH}_3$  under a given set of conditions is 0.35 M/s, then what is the rate of disappearance of  $\text{H}_2$  under the same conditions?



- A) 0.23 M/s
- B) 0.35 M/s
- C) 0.53 M/s
- D) 0.70 M/s
- E) 1.1 M/s

15. The following data were collected for the reaction  $2 \text{A}(\text{g}) + \text{B}(\text{g}) \rightarrow 3 \text{C}(\text{g})$ . What is the rate law for the reaction?

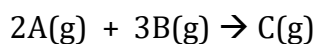
| Experiment | $[\text{A}]_{\text{initial}} (\text{M})$ | $[\text{B}]_{\text{initial}} (\text{M})$ | Rate (M/s) |
|------------|------------------------------------------|------------------------------------------|------------|
| 1          | 2.40                                     | 1.72                                     | 0.68       |
| 2          | 2.40                                     | 3.44                                     | 5.44       |
| 3          | 4.80                                     | 5.16                                     | 36.72      |

- A)  $\text{Rate} = k[\text{A}][\text{B}]$
- B)  $\text{Rate} = k[\text{A}]^3[\text{B}]^2$
- C)  $\text{Rate} = k[\text{A}]^3[\text{B}]$
- D)  $\text{Rate} = k[\text{A}][\text{B}]^3$
- E)  $\text{Rate} = k[\text{A}][\text{B}]^2$

16. The biological half-life of a medication is the time required for the drug to lose half of its pharmacologic activity. The biological half-life of a new medication is 6.0 hours and its decay follows first-order kinetics. How long does it take for medication to decrease to 25% of its original pharmacologic activity?

- A) 0.0479 hours
- B) 0.0834 hours
- C) 2.31 hours
- D) 2.49 hours
- E) 12.0 hours

17. In a series of experiments in the laboratory for the reaction shown below, a student triples the concentration of reactant A while holding the concentration of B constant. The observed effect on the rate is an increase by a factor of 27. What is the order of reaction for A?



- A) zero
  - B) first
  - C) second
  - D) third
  - E) there is not enough information to determine the order of reaction of A
18. According to collision theory, under which circumstances may a collision between molecules *not* result in a chemical reaction?
- A) When collisions occur with insufficient kinetic energy.
  - B) When molecules are not oriented properly.
  - C) When molecules are in the gas phase.
  - D) Both A and B.
  - E) None of A, B, or C. All collisions between molecules result in chemical reactions.
19. Data collected in a laboratory experiment was used to create a graph of  $\ln k$  versus  $1/T$  (T in Kelvin). The slope of the resulting line is  $m$ . Which answer option represents the activation energy for the reaction used to collect the data?
- A)  $E_a/R$
  - B)  $-E_a/R$
  - C)  $mR$
  - D)  $-mR$
  - E)  $\ln A$

20. Which phrase or statement best characterizes activation energy?

- A) The difference in energy of the products and reactants in an overall reaction
- B) The energy at the highest point on a reaction progress diagram minus the energy of the reactants.
- C) The difference in the amount of energy required for breaking bonds in the reactants and forming bonds in the products.
- D) The energy released during the reaction.
- E) The energy required for a reaction to go faster.